clinical failure: fixed-effect meta-analysis (odds ratio)

Comparison: Intervention / Control

Methods

Odds ratios were pooled using a fixed-effect model. A continuity correction of 0.5 was applied to individual study estimates only; the pooled Mantel-Haenszel estimate uses the original counts. Heterogeneity was assessed with Cochran's Q and I^2. Analyses were performed in RC MetaStudio 0.5.1. R packages loaded for this analysis: metafor 5.0.1, meta 8.5-0.

Results

19 studies were included (1097 participants in Intervention, 1080 in Control). The pooled odds ratio was 0.712 (95% CI [0.543, 0.934], z = -2.455, p = .014). The confidence interval excludes 1, with lower odds of the outcome in Intervention than in Control. Heterogeneity: Q = 33.492, df = 18, p = .006. I^2 = 52.2%, which may represent moderate to substantial heterogeneity (Cochrane Handbook ranges).

Table 1. Pooled effect

Effect	Estimate	95% CI	SE	z	p
Odds ratio	0.712	0.543 to 0.934	0.138	-2.455	.014

Table 2. Model fit

Statistic	Value
Studies (k)	19
Participants in Intervention	1097
Participants in Control	1080
Cochran’s Q	33.492
Heterogeneity degrees of freedom	18
Heterogeneity p	.006
I^2	52.2%

Table 3. Study results

Study	Effect	95% CI	Weight
Gonzalez	0.571	0.170 to 1.916	5.6%
Prins	0.436	0.107 to 1.772	5.1%
Giamarellou	0.076	0.004 to 1.441	4.0%
Maller	1.401	0.667 to 2.945	9.6%
Sturm	0.189	0.009 to 4.080	1.6%
Marik	0.395	0.229 to 0.680	33.8%
Muijsken	1.611	0.491 to 5.289	3.4%
Vigano	2.878	0.115 to 71.821	0.0%
Hansen	0.140	0.028 to 0.704	7.8%
De Vries	1.973	0.475 to 8.188	2.3%
Mauracher	0.073	0.004 to 1.311	4.8%
Nordstrom	0.663	0.134 to 3.282	3.0%
Rozdzinski	1.018	0.464 to 2.236	9.9%
Ter Braak	1.356	0.501 to 3.666	5.4%
Tulkens	Not estimable	— to —	0.0%
Van der Auwera	0.187	0.009 to 4.062	1.6%
Klastersky	5.762	1.363 to 24.362	1.3%
Vanhaeverbeek	Not estimable	— to —	0.0%
Hollender	0.316	0.013 to 7.928	0.8%

clinical failure · Binary · Odds Ratio · Comparison: Intervention / Control

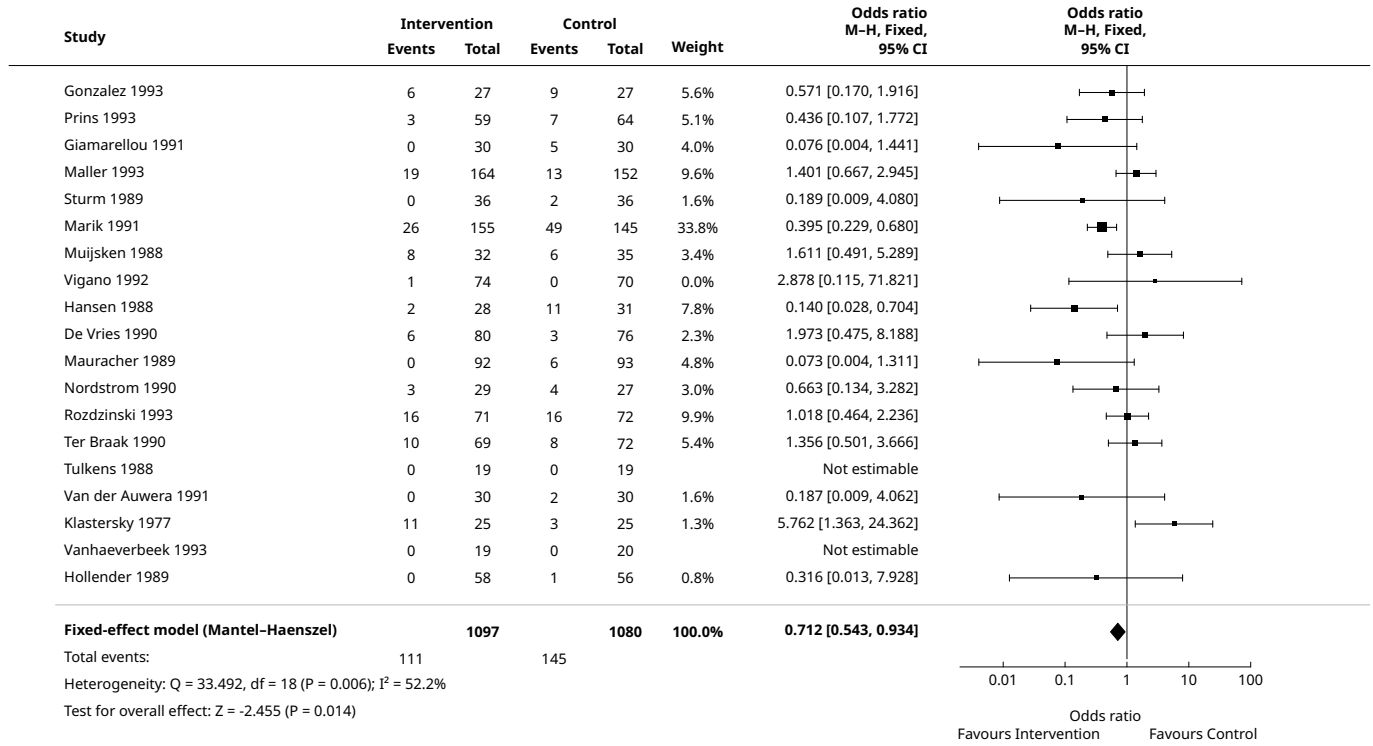

Figure 1. Forest Plot

Supplementary details

Table 4. Full-precision results

Source	Results
Summary — Model Results	Estimate: 0.712 Lower bound (95% CI): 0.543 Upper bound (95% CI): 0.934 p-value: 0.014
Summary — Heterogeneity	Q: 33.492 df: 18 p-value: 0.006 I²: 52.2%
Summary — Test for overall effect	Z: -2.455 p-value: 0.014
Summary — Calculation scale (log)	Estimate: -0.340 Lower bound (95% CI): -0.611 Upper bound (95% CI): -0.069 SE (log scale): 0.138
Weights — Study weights	Study: Gonzalez Effect: 0.571 Lower bound: 0.170 Upper bound: 1.916 Weight (%): 5.6%
Weights — Study weights	Study: Prins Effect: 0.436 Lower bound: 0.107 Upper bound: 1.772 Weight (%): 5.1%
Weights — Study weights	Study: Giamarellou Effect: 0.076 Lower bound: 0.004 Upper bound: 1.441 Weight (%): 4.0%
Weights — Study weights	Study: Maller Effect: 1.401 Lower bound: 0.667 Upper bound: 2.945 Weight (%): 9.6%
Weights — Study weights	Study: Sturm Effect: 0.189 Lower bound: 0.009 Upper bound: 4.080 Weight (%): 1.6%
Weights — Study weights	Study: Marik Effect: 0.395 Lower bound: 0.229 Upper bound: 0.680 Weight (%): 33.8%
Weights — Study weights	Study: Muijsken Effect: 1.611 Lower bound: 0.491 Upper bound: 5.289 Weight (%): 3.4%
Weights — Study weights	Study: Vigano Effect: 2.878 Lower bound: 0.115 Upper bound: 71.821 Weight (%): 0.0%
Weights — Study weights	Study: Hansen Effect: 0.140 Lower bound: 0.028 Upper bound: 0.704 Weight (%): 7.8%
Weights — Study weights	Study: De Vries Effect: 1.973 Lower bound: 0.475

	Upper bound: 8.188 Weight (%): 2.3%
Weights — Study weights	Study: Mauracher Effect: 0.073 Lower bound: 0.004 Upper bound: 1.311 Weight (%): 4.8%
Weights — Study weights	Study: Nordstrom Effect: 0.663 Lower bound: 0.134 Upper bound: 3.282 Weight (%): 3.0%
Weights — Study weights	Study: Rozdzinski Effect: 1.018 Lower bound: 0.464 Upper bound: 2.236 Weight (%): 9.9%
Weights — Study weights	Study: Ter Braak Effect: 1.356 Lower bound: 0.501 Upper bound: 3.666 Weight (%): 5.4%
Weights — Study weights	Study: Tulkens Effect: Not estimable Lower bound: — Upper bound: — Weight (%): 0.0%
Weights — Study weights	Study: Van der Auwera Effect: 0.187 Lower bound: 0.009 Upper bound: 4.062 Weight (%): 1.6%
Weights — Study weights	Study: Klastersky Effect: 5.762 Lower bound: 1.363 Upper bound: 24.362 Weight (%): 1.3%
Weights — Study weights	Study: Vanhaeverbeek Effect: Not estimable Lower bound: — Upper bound: — Weight (%): 0.0%
Weights — Study weights	Study: Hollender Effect: 0.316 Lower bound: 0.013 Upper bound: 7.928 Weight (%): 0.8%

Pooled Odds Ratio: 0.712 (95% CI 0.543 to 0.934).

References

1. Mantel-Haenszel meta-analysis: Mantel, N., & Haenszel, W. (1959). Statistical aspects of the analysis of data from retrospective studies of disease. Journal of the National Cancer Institute, 22(4), 719-748. doi:10.1093/jnci/22.4.719.

2. Common-effect sparse-data estimation: Greenland, S., & Robins, J. M. (1985). Estimation of a common effect parameter from sparse follow-up data. Biometrics, 41(1), 55-68. doi:10.2307/2530643.

3. Implementation reference: Viechtbauer, W. (2010). Conducting meta-analyses in R with the metafor package. Journal of Statistical Software, 36(3), 1-48. doi:10.18637/jss.v036.i03.